Vidyavardhaka Sangha®, Mysore
VIDYAVARDHAKA COLLEGE OF ENGINEERING
Autonomous Institute, Affiliated to Visvesvaraya Technological University, Belagavi
(Approved by AICTE, New Delhi & Government of Karnataka)
Accredited by NAAC with 'A' Grade
Department of Mathematics
VVCE
Phone: +91 821-4276243 , Email: hodmaths@vvce.ac.in
Web: http://www.vvce.ac.in
@vvceofficial

COURSE CONTENT AND PLANNING	
SEMESTER-V	
Course Name: Numerical Methods and its Applications	**Course Code: BMANM561**
No. of Lecture hours / week: 3	**CIE Marks: 50**
No. of Tutorial/Practical hours / week: 0	**SEE Marks: 50**
Total No. of Lecture Hours + Tutorial/Practical hours: 36	**SEE Duration: 3 Hrs**
L: T: P: 3:0:0	**Credits: 3**
COURSE PREREQUISITES: Algebra of Matrices, Elementary Numerical Techniques,	
COURSE OVERVIEW: The objective is to enable the students to apply the knowledge of numerical techniques in various fields of Engineering with applications	
COURSE LEARNING OBJECTIVES The objective is to enable the students to apply the knowledge of advance numerical Mathematics • To find approximate solution of polynomial and simultaneous equation • To evaluate numerical values of differentiation and integration • To evaluate approximate value of eigenvalue and eigenvector of symmetric matrix • To solve differential equation numerically	
COURSE CONTENT	
Module I: Roots of Polynomial Roots of Polynomials: Bairstow's method, Graeffe's Roots Squaring Method. **SLT:** Approximations and round off errors: Significant figures, accuracy and precision, error definitions, roundoff errors and truncation errors.	
Module II: Solution of Linear System - Direct and Iterative Methods Matrix Partition method, Positive Definite matrix and LU- Decomposition, Cholesky decomposition **SLT:** Least Square method for Inconsistent systems.	
Module III: Numerical Differentiation and Numerical Integration Divided difference interpolation, Numerical Differentiation (Newton Forward, backward and divided difference). Romberg integration and Double integral using Simpson's (1/3) rd rule. **SLT:** Maxima and Minima using interpolation	
Module IV: Eigen Values and Eigen Vectors of Symmetric matrices Eigen Values and Eigen Vectors of symmetric matrices using Jacobi and Givens method for symmetricmatrices, Householder's method for symmetric matrices. **SLT:** Rutishauser method for arbitrary matrices.	
Module V: Numerical solution for IVP and BVP **IVP:** Picard's method (first order) **BVP:** Successive Over Relaxation method and Alternating Direction Implicit method. **SLT:** Picard's method (second order)	
Text Books 1. S.S.Sastry, Introductory Methods of Numerical Analysis, PHI, 2005.	

2. Steven C. Chapra, Raymond P. Canale, Numerical Methods for Engineers, Tata McGraw Hill, 4th Ed.,2002.
3. M K Jain, S.R.K. Iyengar, R K jain, Numerical Methods, for Scientific and Engg. Computation, New AgeInternational, 2003.
4. Madhumangal Pal, Numerical Analysis for Scientists and Engineers, Narosa Publishing House, LatestEdition.

Reference Books:

1. Pervez Moin, Fundamentals of Engineering Numerical Analysis, Cambridge, 2010.
2. David C. Lay, Linear Algebra and its Applications, 3rd edition, Pearson Education, 2002.

Course Outcomes: At the end of this course, students will be able to

CO1	**Understand** the numerical methods for solving problems in Calculus and Linear Algebra **(PO-1).**
CO2	**Apply** the concept numerical methods in Calculus and Linear Algebra **(PO-1).**
CO3	**Apply** the concept numerical methods in Calculus and Linear Algebra **(PO-2).**
CO4*	Communicate the concepts of mathematical modeling through presentation.

SEE - Course Assessment Plan

CO	Marks Distribution					Total Marks	Weightage (%)
	Module-1	**Module-2**	**Module-3**	**Module-4**	**Module-5**		
CO1	10	-	-	--	--	10	10%
CO2	10	10	8	20	20	68	68%
CO3	-	10	12	-	-	22	22%
	20	**20**	**20**	**20**	**20**	**100**	100%

Intra-Module Choice for Answering of Questions	
Module - III	**Module - IV**
In other modules there will be no choice for answering	